

156. The equation of a line passing through origin and tangent to a circle $x^2 + y^2 - 2x - 4y = 0$ is
 A. $y = \frac{1}{2}x$ B. $y = \frac{-1}{2}x$ C. $y = 2x + 1$ D. $y = 2x - 1$
157. The distance between the line $3y + 2x - 5 = 0$ and $6y + 4x - 1 = 0$ is
 A. $\frac{9\sqrt{13}}{13}$ B. $\frac{5\sqrt{13}}{13}$ C. $\frac{3\sqrt{3}}{2}$ D. $\frac{9}{\sqrt{52}}$
158. The equation of the ellipse with center at $(1, 4)$, vertices at $(10, 4)$ and $(1, 2)$ is
 A. $(x - 10)^2 + 9(y - 4)^2 = 4$ B. $4(x - 1)^2 + 81(y - 4)^2 = 324$
 C. $9(x - 1)^2 + 4(y - 4)^2 = 1$ D. $2(x - 1)^2 + 9(y - 4)^2 = 4$
159. The equation of parabola satisfying $v(-3, 2)$ and directrix: $x = 1$ is
 A. $(y - 2)^2 = -4(x + 3)$ B. $(x - 2)^2 = -16(y + 3)$
 C. $(y - 2)^2 = -16(x + 3)$ D. $(x - 2)^2 = -4(y + 3)$
160. The intersection point of the circle with equation $(x - 1)^2 + (y + 1)^2 = 25$ with the line $4x - 3y - 7 = 0$ is
 A. $P(-2, -5)$ B. $P(2, 5)$ C. $P(-4, -3)$ D. $P(4, 3)$
161. Let the equation of the circle is $x^2 + y^2 + 2x - 4y - 5 = 0$, Then what is the equation of tangent line at point $(1, 5)$ on the circle?
 A. $2x + y = 11$ B. $x + 2y = 11$ C. $2x + 2y = 11$ D. $x + y = 11$
162. The standard form the equation of a circle with Centre $(-2, 3)$ and tangent to the x-axis is
 A. $(x - 3)^2 + (y - 4)^2 = 9$ B. $(x + 2)^2 + (y - 3)^2 = 9$
 C. $(x + 3)^2 + (y + 4)^2 = 9$ D. $(x - 2)^2 + (y + 3)^2 = 9$
163. Let $x^2 + y^2 + kx = 1$ be equation of a circle for some $k \in R$. what is the radius of the circle if its center is $(2, 0)$?
 A. $\sqrt{5}$ B. 2 C. $\sqrt{2}$ D. 5
164. The line through $(-2, 3)$ and $(2, a)$ is perpendicular to the line through $(2, 5)$ and $(1, 7)$, then what is value of a ?
 A. 4 B. 5 C. 2 D. 3
165. What is the point on the parabola $y^2 = 16x$ whose distance from the focus is 13?
 A. $(9, 12)$ and $(9, -12)$ B. $(12, 9)$ and $(12, -9)$
 C. $(3, 4)$ and $(3, -4)$ D. $(4, 3)$ and $(4, -3)$
166. Which of the following is asymptote of the hyperbola $y^2 - x^2 + 6x - 5 = 0$?
 A. $y = x + 3$ B. $y = -x + 3$ C. $y = -x - 3$ D. $y = x - 4$
167. Which of the following is the distance from center to one of foci of hyperbola $\frac{(y - 2)^2}{20} - \frac{(x + 1)^2}{16} = 1$?
 A. 5 B. 10 C. 6 D. 8
168. What is the point on the line $y = x - 2$ at a distance of $3\sqrt{2}$ units from $(1, -1)$?
 A. $(4, 2)$ B. $(-4, -2)$ C. $(2, 4)$ D. $(3, 4)$
169. The general second degree equation $x^2 + 4xy - 2y^2 - 6 = 0$ represents equation of :
 A. An ellipse B. A parabola C. Hyperbola D. Circle